

Sanchayan Bhowal

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EDUCATION

Stanford University

PhD in Statistics

Stanford, CA

2025-Present

Indian Statistical Institute

Master of Mathematics, Grade: 97.7%

Bangalore, India

2023-2025

Project Advisor: [Dr. Bhaswar Bhattacharyya](#) and [Dr. Somabha Mukherjee](#)

Bachelor of Mathematics, Grade: 94.5%

2020-2023

RESEARCH

- **Bhowal, S.**, Chatterjee, A., & Mukherjee, S. (2026). Scaling Limits for Ising Models on Inhomogeneous Random Graphs and Applications 
- Mukherjee, S., **Bhowal, S.**, Chatterjee, A., & Bhattacharyya, B. (2026). Ising Models on Inhomogeneous Random Graphs: Inference, Local Asymptotic Maximality, and Limit of Experiments 
- Bhattacharyya, B., **Bhowal, S.**, Das, K., Eslava, L. & Karmakar, S. Thresholds and Fluctuations in Random Multiplex Networks. Accepted at *Canadian Mathematical Bulletin* 
- **Bhowal, S.** & Mukherjee, S. Rates of Convergence of the Magnetization in the Tensor Curie-Weiss Potts Model. *Journal of Statistical Physics* 192, 2 (2025). 
- **Bhowal, S.** & Mukherjee, S. (2023). Limit Theorems and Phase Transitions in the Tensor Curie-Weiss Potts Model. *Information and Inference: A Journal of IMA*. 
- **Bhowal, S.**, Samanta, R., Ray, A., Bhattacharyya, S. & Hens, C. (2023) Exploring the potential of collective learning to reduce foraging time. *Chaos, Solitons & Fractals*, 168, 113123 

EXPERIENCE

Big Data Summer Institute, University of Michigan

June 2023 – July 2023

[Dr. Snigdha Panigrahi](#) & [Dr. Johann Gagnon-Bartsch](#)

In this project we compared various feature selection methods. We created two separate models with the best feature selection method for genomics and imaging data to predict survival time in LGG Patients. We also employed stability selection in order to ensure robustness.

SN Bhatt Memorial Fellowship Program

May 2022-July 2022

[Dr. Anirban Basak](#) & [Dr. Riddhipratim Basu](#)

I worked on Cutoff Phenomenon in Markov Chains. This included reading from the book *Markov Chains and Mixing Times*. I also read the paper *Total Variation Cutoff in Birth Death chains* by Ding, Lubetzky and Peres.

TALKS

Limit Theorems of Estimators in The Tensor Curie Weiss Potts Model

2024

Invited talk at [Bangalore Probability Seminar](#)

What is Cutoff?

2024

[Math Club ISI Bangalore](#)

A tour of Markov chains

2022

[Math Club ISI Bangalore](#)

An Unconventional Approach to Groups

2021

[ISI Students Seminar](#)

AWARDS AND FELLOWSHIPS

- P.C. Panesar Gold Medal for outstanding performance in M. Math Programme (2023-25)
- Regional prize Madhava Mathematics Competition (2023)
- National award ISLP Poster Competition (2022)
- KVPY fellowship (2020- 2025)
- JBNSTS Scholarship (2018-2020)

TEACHING EXPERIENCE

STATS 60: Introduction to Statistical Methods: Precalculus	Summer 2026
STATS 60: Introduction to Statistical Methods: Precalculus	Winter 2026
STATS 110: Introduction to Statistics for Engineering and the Sciences	Fall 2025

CONFERENCES

Geometry, Probability, and Algorithms Organizers: Akash Kumar, Anand Louis and Piyush Srivastava	May 12-16 2025
High Dimensional Expanders and Error Correcting Codes Organizers: Irit Dinur, Venkat Guruswami and Prahladh Harsha	May, 1-9 2025
Mathematical Foundations of Network Models and their Applications Organizers: Louigi Berry, Siva Athreya, Shankar Bhamidi, Serte Donderwinkel, Soumik Pal	Dec, 9-21 2024
International Colloquium on Randomness, Geometry, and Dynamics Organizers: TIFR Mumbai and IISER Pune	Jan, 1-5 2024
Topics in High Dimensional Probability Organizers: Anirban Basak and Riddhipratim Basu	Jan, 2-13 2023
Vijyoshi Organizer: IISc Bangalore for KVPY fellows	Dec, 7-9 2019

SKILLS

Programming Languages: C/C++, Python, R, JavaScript, HTML/CSS, $\text{\LaTeX}$
Languages: Bengali(Native), English, Hindi, Assamese

PROFESSIONAL SERVICES

Reviewer: Annals of Applied Probability, Journal of Physics A: Mathematical and Theoretical, Sankhya B: The Indian Journal of Statistics.

Organizational Roles

- Student Coordinator, **Math Club** ISI Bangalore, 2022-2025
- Core Team Member, LIMIT, an online mathematics competition, 2021-2025

Last Updated on August 13, 2026